Beyond Model Benchmarks: A Framework for Evaluator Readiness in Human-Generative AI Interaction

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Abstract

Human evaluation is central to benchmarking culturally competent and socially responsible generative artificial intelligence (GenAI), yet evaluation protocols often characterize the model and task more carefully than the humans whose judgments define benchmark outcomes. This position paper introduces *evaluator readiness* as a complementary construct to AI literacy and community or cultural expertise. We distinguish awareness of AI limitations-knowing that GenAI may hallucinate, reproduce bias, omit context, or require expert validation-from readiness to evaluate AI outputs through observable verification, judgment, and human-agency behaviors. This distinction is motivated by prior survey evidence from data science education: more than 70% of students recognized major GenAI limitations, while only 40.6% reported confidence explaining AI-generated outputs. We propose an Evaluator Readiness Framework (ERF) for analyzing human-GenAI interaction logs and reflections. ERF separates *interaction competency* (prompt quality and iterative engagement) from *evaluator readiness* (limitation awareness, metacognitive monitoring, verification, evaluative judgment, and human agency). We describe behaviorally anchored coding levels, a profile-based rather than a one-dimensional scoring strategy, and a validation agenda using inter-rater reliability, survey-behavior correspondence, and interaction traces. We argue that community-informed evaluation of culturally competent GenAI requires attention to both *who* evaluates and *how prepared* evaluators are to interrogate AI outputs. The framework provides a basis for more transparent, human-centered evaluation protocols in education and, more broadly, for developing culturally grounded benchmarks.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models**; • **Computing methodologies** → **Artificial intelligence**; • **Social and professional topics** → *Computing education.*

Keywords

AI literacy, evaluator readiness, GenAI, human oversight, human-centered evaluation, verification.

1 Introduction

Evaluation of generative AI increasingly depends on human judgment. This dependence matters most for culturally competent AI, where benchmark items may involve social norms, lived experience, language variation, representational harms, or context that cannot be reduced to a single objective label. Recent work shows that annotators are not fungible: demographic background and lived experience can systematically shape judgments in subjective NLP tasks [2, 6, 7]. Cultural benchmarks similarly rely on human construction and verification to ensure that evaluation items reflect culturally grounded knowledge rather than translated or decontextualized approximations [1, 8].

These developments make evaluator composition visible, but a second measurement problem remains underdeveloped: *Are evaluators prepared to critically interrogate GenAI output?* Community membership, cultural expertise, or demographic representation may be indispensable for culturally grounded evaluation, but none alone guarantees the ability to recognize hallucinated evidence, test analytical claims, identify unsupported assumptions, or decide when an apparently fluent response should be revised or rejected. Conversely, technical expertise does not substitute for cultural or community knowledge. Valid human-centered evaluation therefore requires attention to both evaluator context and evaluator readiness.

AI literacy frameworks already include the capacity to critically evaluate AI [5], but educational measurement often operationalizes literacy through familiarity, perceptions, confidence, or stated awareness. A recent survey of 119 statistics and data science students found that at least 69% recognized key limitations of GenAI, including biased or inaccurate output and the need for expert validation, while only 40.6% reported confidence explaining AI-generated outputs [3]. This discrepancy motivates a distinction between *awareness of limitations* and *evaluator readiness*: knowing that AI can fail is not the same as demonstrating the capacity to verify, challenge, and take responsibility for AI-assisted decisions.

This paper contributes: (1) a conceptual separation of interaction competency, limitation awareness, and evaluator readiness; (2) a behaviorally anchored framework for coding evaluator readiness from human–GenAI interaction logs and reflections, with its awareness dimension derived directly from a validated survey instrument's item structure; (3) a preliminary inter-rater agreement analysis applying ERF's coding scheme to real student interaction logs; and (4) a research agenda connecting evaluator readiness to community-informed and culturally grounded AI evaluation.

2 From AI Literacy to Evaluator Readiness

2.1 Why interaction quality is not enough

Effective interaction with GenAI is often inferred from sophisticated prompting or iterative dialogue. Yet a well-constructed prompt can still be followed by uncritical acceptance of an incorrect response. Likewise, a comparatively simple prompt can initiate strong verification if the user checks claims independently and rejects unsupported output. We therefore separate two layers:

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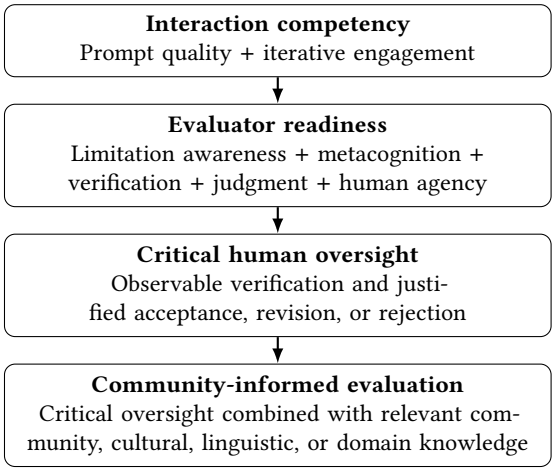


Figure 1: ERF positions evaluator readiness as a behavioral prerequisite for critical human oversight; culturally grounded evaluation additionally requires relevant community or contextual expertise.

Interaction competency. The ability to elicit useful AI assistance through clear prompts and purposeful iteration.
Evaluator readiness. The demonstrated capacity to monitor limitations, verify outputs, exercise judgment, and retain epistemic responsibility for the final decision.

This distinction complements recent work such as RelianceScope, which characterizes students' engagement in help-seeking and response use [4]. ERF focuses on a different question: not simply *how much* a person relies on AI, but whether the interaction contains behavioral evidence that the person can serve as a critical evaluator of AI output.

2.2 The Evaluator Readiness Framework

ERF adapts a behaviorally anchored rubric originally designed for student AI-interaction logs. The original dimensions assess prompt quality, interaction/iteration, metacognitive engagement, verification/evaluation, and over-reliance. We retain these behavioral foundations, reinterpret over-reliance positively as *human agency*, and add two dimensions required to connect behavior to the earlier survey construct: *limitation awareness* and *evaluative judgment*.

Table 1 summarizes the resulting framework. We recommend ordinal scores from 1-4 within each dimension, with explicit behavioral anchors rather than global judgments. Importantly, ERF is a *profile*, not yet a validated additive scale. Prompt quality and iteration constitute interaction competency; the remaining five dimensions characterize evaluator readiness.

3 Applying ERF to Interaction Logs and Reflections

3.1 Unit of analysis and evidence sources

The natural unit of analysis is an *interaction episode*: a prompt, the AI response, subsequent user actions or follow-up prompts, and, when available, the artifact produced or revised. Structured reflections provide a second evidence stream for metacognitive monitoring and accountability. Surveys provide a third stream capturing perceived awareness or confidence. Together, these sources separate three levels of evidence:

$$\text{self-report} + \text{interaction behavior} + \text{reflection} \Rightarrow \text{evaluator-readiness profile.}$$

For example, a student who reports strong awareness that GenAI can hallucinate may nevertheless accept generated code or statistical interpretation without checking. ERF would record high *reported awareness* but low observed verification and judgment. Conversely, a cautious student may report modest confidence yet systematically reproduce calculations, inspect assumptions, and reject an incorrect response. Such cases motivate a calibration perspective rather than equating confidence with competence.

3.2 Pilot assessment of ERF coding consistency

To assess whether ERF-aligned behaviors can be coded consistently, we conducted a small pilot using ten real student interaction logs from AI-assisted data analytics coursework. The episodes were student-shared conversation transcripts from sessions with ChatGPT, Gemini, and Copilot. Each episode was scored on the five dimensions in the original interaction rubric: prompt quality, interaction pattern, metacognition, verification, and over-reliance (with over-reliance reverse-coded here as human agency). A human coder, labeled *Student Tester #1* for anonymous review, and one or more LLM-assisted raters scored overlapping subsets. Because coverage was incomplete, we do not pool all rater pairs or interpret the pilot as a formal inter-rater reliability study. Table 2 reports the most complete paired comparison, Student Tester #1 and ChatGPT ($n = 8$).

The descriptive indices are encouraging but nonuniform. Rank concordance was highest for interaction pattern ($\rho = .88$) and metacognition and human agency ($\rho = .84$), whereas prompt quality was lower ($\rho = .61$). Exact agreement varied more widely, from 25% for prompt quality to 88% for verification, while within-one-point consistency was at least 88% for every dimension. The combination of high exact agreement but more moderate rank concordance for verification also cautions against interpreting any single statistic in this small, tie-prone ordinal sample. Rather than ranking dimensions as more or less reliable, the pilot suggests that the behavioral anchors are operationally codable while motivating dimension-specific reliability analysis with a larger multi-human design.

LLM-assisted coding is treated here as a supplementary reproducibility stress test, not as ground truth. Human-LLM concordance cannot establish coding validity because the model's own evaluative reasoning is not independently verified; this methodological tension itself illustrates ERF's distinction between fluent output and grounded evaluative judgment.

3.3 Staged coding and validation

We therefore propose a three-stage validation design. **Stage 1 (current)** provides construct grounding and pilot operationalization: limitation awareness is motivated by the previously evaluated survey construct, while the remaining behavioral dimensions are instantiated through the interaction rubric and subjected to the preliminary consistency check above. This does not validate the

Table 1: Evaluator Readiness Framework (ERF) for coding human-GenAI interaction evidence.

Layer	Dimension	Low behavioral anchor	High behavioral anchor
Interaction competency	Prompt quality	Vague or minimal request with little context or purpose.	Specific, contextualized prompt with goals, constraints, comparison, or purposeful refinement.
Interaction competency	Iterative engagement	Single-turn or minimal follow-up.	Targeted follow-ups that refine the problem, request clarification, or redirect the AI based on prior output.
Evaluator readiness	Limitation awareness	Treats output as inherently reliable; recognizes no relevant limitation.	Identifies or anticipates a task-relevant limitation (e.g., hallucination, bias, missing context, uncertainty, privacy) and incorporates it into evaluation.
Evaluator readiness	Metacognitive monitoring	No evidence of reflection on reasoning, uncertainty, or strategy.	Articulates reasoning and uncertainty; reflects on why output should or should not be trusted and on the role of AI in the reasoning process.
Evaluator readiness	Verification	Accepts output without checking or performs only surface plausibility checking.	Systematically cross-checks claims, code, calculations, assumptions, sources, or results using independent evidence.
Evaluator readiness	Evaluative judgment	Accepts or edits output without a reasoned accept/modify/reject decision.	Makes and justifies an explicit accept/modify/reject decision using evidence, assumptions, domain knowledge, or statistical reasoning.
Evaluator readiness	Human agency	AI substantially directs the task and the user contributes little independent reasoning.	User directs the process, retains task ownership, and can explain or defend the final conclusion independently of the AI response.

Table 2: Pilot coding consistency between Student Tester #1 (human coder) and ChatGPT ($n = 8$ paired episodes). Spearman's ρ is reported as rank concordance, not as an agreement coefficient.

Dimension	ρ	Exact	Within ±1
Prompt quality	.61	25%	88%
Interaction pattern	.88	50%	100%
Metacognition	.84	75%	100%
Verification	.68	88%	88%
Human agency	.84	62%	100%

Table 3: Illustrative evaluator-readiness profiles.

Profile	Behavioral interpretation
Fluent adopter	Strong prompting/iteration, but little verification or independent judgment.
Aware but unprepared	Recognizes AI limitations conceptually, but does not translate awareness into checking or revision.
Critical supervisor	Verifies claims, interrogates assumptions, and justifies acceptance, modification, or rejection.
Contextual evaluator	Critical-supervisor behaviors combined with relevant cultural, community, linguistic, or domain knowledge.

new behavioral limitation-awareness dimension or ERF as a whole. **Stage 2** will use a systematically sampled interaction set and two or more trained human coders, with ordinal reliability estimated using an appropriate agreement coefficient such as ordinal Krippendorff's α; LLM-assisted ratings should remain supplementary and independently checked. **Stage 3** will test construct validity: survey limitation-awareness scores should align more strongly with observed limitation recognition than with prompt quality, while verification and judgment should covary yet remain distinguishable from interaction competency. Reflection evidence can additionally test whether users can explain how and why they verified, modified, or rejected AI output.

We explicitly do *not* recommend summing all dimensions into a single readiness score before dimensionality and reliability are established. A profile representation preserves diagnostically important patterns, such as sophisticated prompting paired with weak verification or strong risk awareness paired with low evaluative agency.

3.4 Interpretive profiles

The profile representation is designed to reveal combinations that a single total score would conceal. Table 3 illustrates four analytically distinct patterns. These are conceptual profiles rather than empirical classifications at this stage.

4 Implications for Community-Informed AI Evaluation

REACH-AI foregrounds culturally grounded benchmarks and community-informed assessment. In such settings, evaluator diversity is necessary but not sufficient. Prior annotation research shows that demographic and experiential differences can shape judgments and that disagreement may encode meaningful perspectives rather than noise [2, 6, 7]. ERF adds a complementary question: what evidence shows that evaluators can critically interrogate the AI response they are rating?

This distinction avoids two problematic assumptions. First, cultural or demographic identity should not be treated as a proxy for technical evaluation competence. Second, technical competence

should not be treated as a proxy for cultural understanding. A credible evaluation protocol may therefore need to document both (a) the evaluator's relevant community, linguistic, cultural, or domain context and (b) the evaluator's readiness to recognize uncertainty, verify claims, articulate reasons, and resist unwarranted model authority.

For culturally grounded benchmark development, ERF suggests several reporting practices: describe evaluator preparation and calibration; document verification resources and procedures; report dimension-specific coding reliability rather than assuming uniform coder performance; preserve justified disagreement rather than automatically collapsing it to majority vote; and report whether evaluators were asked to explain acceptance, revision, or rejection decisions. These practices could make the human side of benchmarking more auditable and help distinguish model failure from evaluator uncertainty or insufficient task preparation.

5 Limitations and Research Agenda

ERF remains a proposed measurement framework rather than a validated psychometric instrument. The pilot demonstrates initial practical applicability, not reliability or construct validity: it includes only ten episodes, incomplete rater coverage, one human coder, and LLM-assisted comparison ratings. Human-LLM concordance cannot substitute for multi-human inter-rater reliability, and the small ordinal sample is sensitive to ties and restricted score variation. Self-reported survey results motivate the awareness-readiness distinction but do not establish behavioral readiness. Moreover, the present student interaction context does not itself measure cultural competence unless tasks explicitly involve culturally situated reasoning, representation, fairness, or bias. Model-assisted coding of interaction records also raises data-governance questions that future protocols should address explicitly.

These limitations define the next research steps. Immediate work will expand systematic sampling of student GenAI interaction logs and structured reflections, use multiple trained human coders to estimate ordinal reliability, refine coding anchors, and test correspondence between survey awareness and observed verification. Subsequent studies should apply ERF to culturally situated evaluation tasks and examine whether evaluator-readiness profiles affect ratings, disagreement, and benchmark conclusions across community and institutional contexts. The broader goal is not to gate community participation, but to make evaluation protocols transparent about the preparation, reasoning, and uncertainty of all evaluators.

6 Conclusion

Human judgments are measurements, not neutral endpoints. As AI evaluation becomes more community-informed and culturally grounded, validity depends on both the perspectives evaluators bring and their readiness to critically supervise GenAI. ERF operationalizes this readiness through observable limitation awareness, metacognition, verification, judgment, and human agency while keeping these behaviors distinct from prompt sophistication. The preliminary coding study suggests that these dimensions can be operationalized in real interaction traces while also showing why dimension-specific reliability must be established before scores are treated as stable measurements. ERF thus offers a concrete path from self-reported AI literacy toward auditable evidence of human oversight and a research agenda for incorporating evaluator readiness into culturally competent GenAI benchmarking.

GenAI Usage Disclosure

The empirical materials include student-shared interactions with ChatGPT-5.4 , Gemini, and Copilot produced during AI-assisted coursework. OpenAI's ChatGPT was used as a supplementary coder in the preliminary coding-consistency analysis reported in Section 3.2 and was not treated as ground truth. ChatGPT also assisted with literature discovery and manuscript organization. Grammarly was used for language refinement. The authors reviewed the coding outputs, empirical summaries, cited sources, factual claims, and final manuscript and retain full responsibility for the analysis and interpretation.

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References

[1] Yu Ying Chiu, Liwei Jiang, Bill Yuchen Lin, Chan Young Park, Shuyue Stella Li, Sahithya Ravi, Mehar Bhatia, Maria Antoniak, Yulia Tsvetkov, Vered Shwartz, and Yejin Choi. 2025. CulturalBench: A Robust, Diverse and Challenging Benchmark for Measuring LMs' Cultural Knowledge Through Human-AI Red-Teaming. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (Eds.). Association for Computational Linguistics, Vienna, Austria, 25663–25701. doi:10.18653/v1/2025.acl-long.1247

[2] Eve Fleisig, Rediet Abebe, and Dan Klein. 2023. When the Majority is Wrong: Modeling Annotator Disagreement for Subjective Tasks. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 6715–6726. doi:10.18653/v1/2023.emnlp-main.415

[3] Abeer M. Hasan and Sayed A. Mostafa. 2026. Perceptions and Utilization of GenAI Tools Among Data Science Students and Faculty. *J. Data Sci.* 24, 4 (2026), 1026–1042. doi:10.6339/26-JDS1233

[4] Hyoungwook Jin, Minju Yoo, Jieun Han, Zixin Chen, So-Yeon Ahn, and Xu Wang. 2026. RelianceScope: An Analytical Framework for Examining Students' Reliance on Generative AI Chatbots in Problem Solving. In *Proceedings of the Thirteenth ACM Conference on Learning @ Scale*. Association for Computing Machinery, 136–147. doi:10.1145/3774398.3811612

[5] Duri Long and Brian Magerko. 2020. What is AI Literacy? Competencies and Design Considerations. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, 1–16. doi:10.1145/3313831.3376727

[6] Aida Mostafazadeh Davani, Mark Díaz, and Vinodkumar Prabhakaran. 2022. Dealing with Disagreements: Looking Beyond the Majority Vote in Subjective Annotations. *Transactions of the Association for Computational Linguistics* 10 (2022), 92–110. doi:10.1162/tacl_a_00449

[7] Jiaxin Pei and David Jurgens. 2023. When Do Annotator Demographics Matter? Measuring the Influence of Annotator Demographics with the POPQUORN Dataset. In *Proceedings of the 17th Linguistic Annotation Workshop (LAW-XVII)*. Association for Computational Linguistics, 252–265. doi:10.18653/v1/2023.law-1.25

[8] Ishaan Watts, Varun Gumma, Aditya Yadavalli, Vivek Seshadri, Manohar Swaminathan, and Sunayana Sitaram. 2024. PARIKSHA: A Large-Scale Investigation of Human–LLM Evaluator Agreement on Multilingual and Multi-Cultural Data. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 7900–7932. doi:10.18653/v1/2024.emnlp-main.451